Domain Adaptive Land-Cover Classification via Local Consistency and Global Diversity

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***Abstract*— Unsupervised domain adaptive land-cover classification has recently gained more and more attention. Unsupervised domain adaptation (UDA) aims to learn a model from the annotated source data and the unlabeled target data that can perform well on the target domain. The existing UDA frameworks based on adversarial training and self-training methods have boosted this field a lot. However, these methods almost all originate from the computer vision field, and they ignore the very nature of high-resolution remote sensing (HRS) images. The core insight of this article is that a good land-cover classification result always has strong local consistency and good global diversity, which makes it possible to construct a metric representing the properties of good land-cover mapping, to improve the existing UDA algorithms. First, based on this finding, we prove that local consistency and global diversity can be measured by the Frobenius norm and nuclear norm, respectively. Second, we propose a novel local consistency and global diversity metric (LCGDM), which can be easily integrated into the existing UDA frameworks. Finally, the experiments conducted on the LoveDA dataset prove the validity of the proposed metric, which can not only improve the overall land-cover mapping but also the categorywise prediction.**

***Index Terms*— Consistency and diversity, high-resolution remote sensing (HRS) images, land-cover classification, unsupervised domain adaptation (UDA).**

I. Introduction

LAND-COVER classification, which is also named semantic segmentation from the perspective of the computer vision field, is aimed at distinguishing high-resolution remote sensing (HRS) images with pixelwise precision and has long been a significant task for many applications in the remote sensing (RS) community [1], [2], [3], [4]. With the advances in Earth observation technology, a large number of HRS images containing more detailed spatial information have become available, which makes better land-cover classification possible. However, some severe problems do exist. On the one hand, to achieve refined land-cover mapping in a specified scene, the existing methods based on deep learning techniques require massive labeled HRS images of the scene, which are very time-consuming to annotate manually. On the other hand, the data distributions of different geographic areas (e.g., urban and rural) can be diverse, so it is extremely difficult to apply a deep neural network (DNN) trained on the source scene to the target domain directly due to the large discrepancy between data distributions of the two domains. These problems have greatly limited the development of HRS image land-cover classification and its applications.

A considerable amount of the literature has been published to address these issues [5], [6], [7], [8], [9], [10]. A promising approach among these studies is unsupervised domain adaptation (UDA) [11], which aims to make use of the labeled source data and the unlabeled target data to learn a discriminant function that can perform well in the target domain when there is a shift between the two domains. More recently, UDA semantic segmentation based on DNNs [12], [13], [14] has been proposed. The methods based on this approach can be mainly divided into adversarial training (AT) methods [15], [16], [17], [18] and self-training (ST) methods [19], [20], [21], [22]. The former methods minimize a series of adversarial losses to align the source and target feature distributions, and adopt a discriminator to distinguish them. Meanwhile, the latter methods iteratively fine-tune the model by using both the labeled source domain data and the generated reliable pseudolabels from the target domain and, thus, achieve alignment between the source and target domains. Furthermore, some researchers [23], [24], [25] have attempted to combine AT and ST methods, which is an approach that has achieved great improvements in UDA semantic segmentation.

Despite the success of these AT and ST methods originating from the computer vision field, the existing methods always aim to improve the AT or ST training framework by changing the training process (e.g., more complex pseudolabels' construction and more refined distribution alignment), but they ignore the natural properties of HRS images that can very much enhance the prediction performance. Some natural questions come up. Are there any UDA methods designed specifically for RS tasks? Are there any metrics representing the prior knowledge of good land-cover prediction to improve the existing UDA algorithms for RS? To date, few studies have investigated these questions.

Unlike the existing solutions, we do not intend to propose a novel AT or ST framework but instead focus on the very nature of HRS images. The core insight of this article is based on

Manuscript received 29 November 2022; revised 7 March 2023; accepted 25 March 2023. Date of publication 6 April 2023; date of current version 19 April 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903404 and in part by the National Natural Science Foundation of China under Grant 42171336. *(Corresponding authors: Ailong Ma; Chenyu Zheng.)*

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Digital Object Identifier 10.1109/TGRS.2023.3265186

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the statistical fact that a good land-cover classification result should have strong local consistency [26] and good global diversity [27]. That is, on the one hand, when we focus on a small region of an HRS image, the number of categories of objects is relatively low, and we hope to generate consistent and highly confident predictions on these small local regions. On the other hand, when we consider the whole HRS image, there are always lots of objects from different categories, so we are also supposed to keep the global diversity of the segmentation result.

In this article, to improve the existing UDA methods' performance on HRS imagery, we propose a novel metric that represents the local consistency and global diversity of HRS imagery. The main contributions of our work are listed as follows.

First, to measure the local prediction consistency, several local prediction matrices are cropped from the whole high-resolution output. A common solution to strengthen the local prediction consistency is minimizing the Shannon entropy [28]. We chose the Frobenius norm, an equivalent version of Shannon entropy, in our setting, as the metric for the local consistency to maintain consistency with the following global diversity metric and reduce computational consumption.

Second, to measure the global diversity, a global matrix is defined, and the vector whose elements represent the occurrence probability of the corresponding category in each region is considered as the related row vector of the global matrix. In fact, the rank of the global matrix can be used to measure the global diversity of the prediction. However, this observation is not practically useful on its own. To solve this problem, the nuclear norm is used as the metric of global diversity, which is a natural convex relaxation of rank [29] and can be calculated quickly by using singular value decomposition (SVD).

Finally, the local consistency and global diversity metric (LCGDM) based on the above theoretical findings is proposed. The local consistency and global diversity of the land-cover classification result can be improved by maximizing it. The LCGDM can be integrated into the optimization process of the existing UDA algorithms with only slight extra computational effort. The experimental results obtained on the LoveDA land-cover dataset [30] demonstrate that adding the proposed LCGDM to the existing UDA methods can enhance the performance in the target domain by a large margin.

The rest of this article is organized as follows. Section II starts by reviewing some of the related works. In Section III, the proposed LCGDM is described, in addition to its properties and application in the UDA training framework. In Section IV, the empirical results on the LoveDA dataset [30] are reported. In Section V, we discuss the implications of the proposed LCGDM. Finally, our conclusions are drawn in Section VI.

II. Related Work

A. UDA Based on Adversarial Training

AT methods minimize a series of adversarial losses to align the source and target feature distributions, and adopt a discriminator to distinguish domain-invariant features. Ganin and Lempitsky [31] first promoted the emergence of features that are discriminative for the main learning task on the source domain and invariant with respect to the shift between domains, which performed well in image classification experiments. For dense prediction tasks, Tsai et al. [15] constructed a multilevel adversarial network (AdaptSeg) to effectively perform output space domain adaptation at different feature levels. Luo et al. [17] introduced a category-level adversarial network (CLAN), which takes a close look at the category-level joint distribution and aligns each class with an adaptive adversarial loss. Wang et al. [16] further proposed a fine-grained adversarial learning strategy (FADA) for class-level feature alignment while preserving the internal structure of the semantics across domains. More recently, some algorithms based on AT methods have been adopted for land-cover classification. Ji et al. [18] adopted a generative adversarial network (GAN)-based domain adaptation [32] for land-cover classification by fully aligning the image space, feature space, and output space. Deng et al. [33] proposed a scale-aware adversarial learning framework to perform joint cross-location and cross-scale land-cover classification. Despite some promising results having been achieved, most of these methods ignore the prior knowledge of good land-cover classification results.

B. UDA Based on Self-Training

ST methods improve the performance in the target domain by alternatively generating pseudolabels on the target data and retraining the model with these labels. Zou et al. [20] proposed a novel UDA framework based on an iterative class-balanced ST (CBST) procedure and achieved good performance on semantic segmentation tasks. Lian et al. [19] adopted self-motivated pyramid curriculum domain adaptation (PyCDA) to facilitate the adaptation of semantic segmentation neural networks. Mei et al. [21] further proposed an instance adaptive ST (IAST) framework for UDA in the task of semantic segmentation by generating pseudolabels with an instance adaptive selector. More recently, some methods inspired by ST methods have been adopted for land-cover classification. Zhang et al. [25] addressed the intradomain discrepancy in the target domain by using an adversarial ST method. Wu et al. [34] explicitly aligned category features to learn shared domain-invariant feature representations, which enhances the ability of model generalization. Wang et al. [22] adopted multiscale pseudolabeling to reduce the scale divergence brought by the spatial resolution inconsistency in cross-sensor tasks. However, most of these methods ignore the very nature of land-cover classification predictions.

C. Toward Consistency and Diversity

Apart from these AT and ST methods, some researchers have already paid attention to the consistency and diversity of the prediction. To enhance the reliability of the prediction on the target domain, a straightforward approach is minimizing the Shannon entropy. Long et al. [35] exploited the

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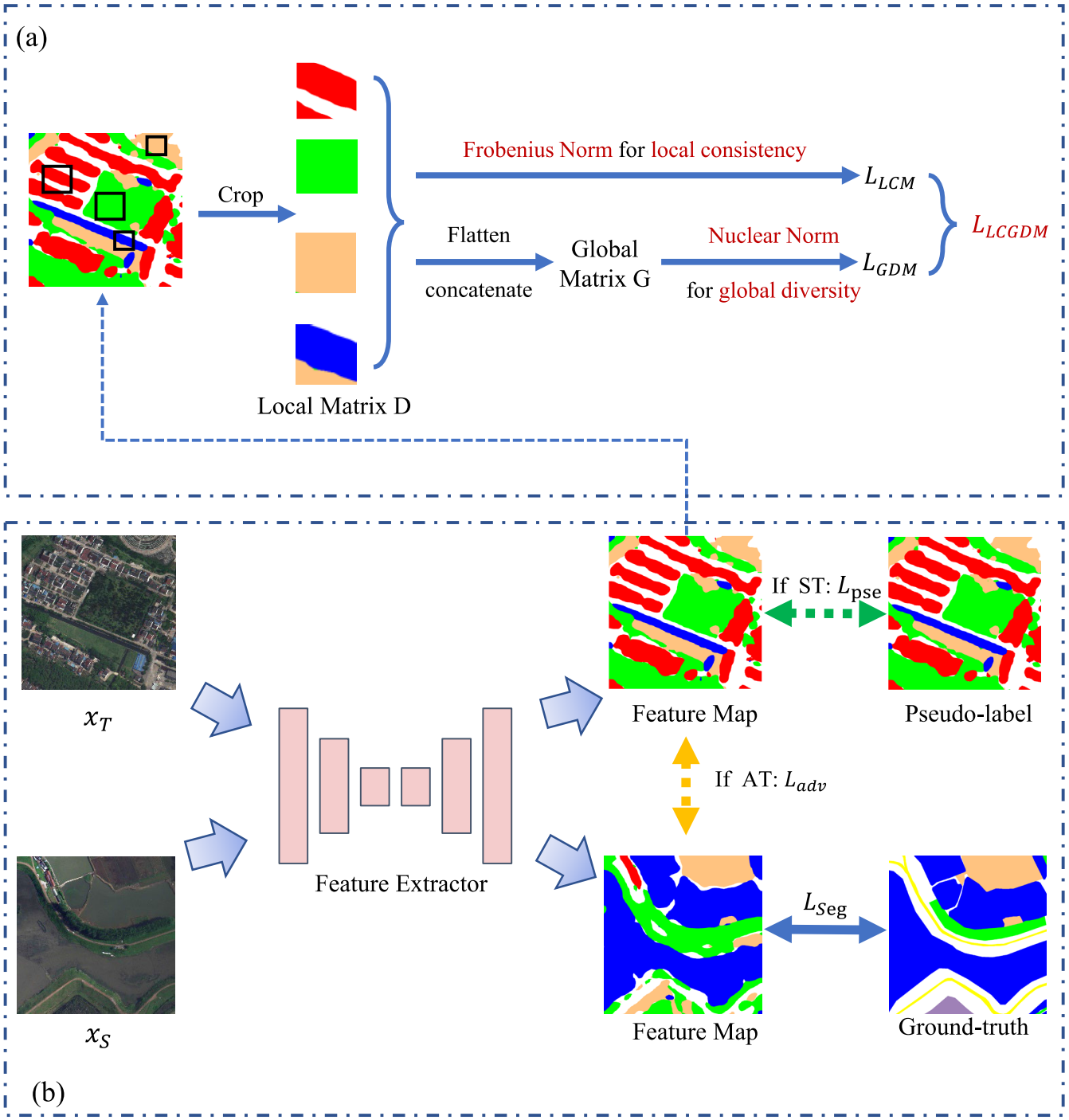


Fig. 1. Overview of the proposed AT/ST training framework based on the auxiliary LCGDM loss function. (a) Calculation process for $\mathcal{L}_{\text{LCGDM}}$ corresponding to Algorithm 1 (proposed $\mathcal{L}_{\text{LCGDM}}$ loss function). (b) Compatible AT/ST training framework. x_T is sampled from the target domain, and x_s is an example from the source domain. $\mathcal{L}_{\text{Seg}}$ is obtained based on the ground truth of x_S and the feature map of x_S. If the basic framework is AT, then $\mathcal{L}_{\text{adv}}$ will be computed by combining the feature map of x_S and the feature map of x_T. If the basic framework is ST, then $\mathcal{L}_{\text{pse}}$ will be computed based on the feature map of x_T and the pseudolabel for it. Methods that use both AT and ST techniques are also included in this framework.

entropy minimization principle on a class-conditional distribution to encourage low-density separation between classes. Vu et al. [26] addressed the task of UDA in semantic segmentation based on minimizing the entropy of the pixelwise predictions directly. Liang et al. [36] further improved the prediction consistency by minimizing the cross-entropy loss with pseudolabels. However, minimizing the Shannon entropy can lead to the reduction of diversity [37] and transferability [38].

In order to promote the diversity of predictions, many solutions have been proposed based on imbalanced learning [39]. For example, Zou et al. [20] utilized confidence scores normalized in a classwise manner to select and generate pseudolabels to keep the diversity of the results; Zou et al. [27] proposed label regularization and model regularization to generate soft pseudolabels and network outputs; and Liang et al. [36] increased prediction diversity by

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minimizing the Kullback–Leibler (KL) distance [40] between the outputs' classwise distribution in the target domain and the uniform distribution. However, they only focus on image classification tasks.

Cui et al. [41] made use of the nuclear norm to improve the consistency and diversity at the same time for the task of UDA in image classification. Although the mathematical tools that we use in the proposed approach are similar, we focus on the problem which is totally different. On the one hand, the LCGDM is proposed for semantic segmentation instead of image classification. On the other hand, as far as we know, we are the first to use these theoretical tools to model the very nature of the predictions in HRS images.

III. Method

In this section, we propose a novel algorithm based on the core insight that a good land-cover prediction result should have strong local consistency and good global diversity. First, we find that the local consistency and global diversity can be measured by the Frobenius norm and the nuclear norm of some matrices built upon the land-cover result. Second, a computational-friendly loss function based on the above theoretical findings is proposed. Finally, the proposed loss function can be conveniently integrated into the minibatch optimization process of the existing UDA algorithms based on ST and AT. The overview of our method is presented in Fig. 1.

A. Measuring the Local Consistency With the Frobenius Norm

We start by looking at the prediction output for an HRS image in the target domain. The number of categories is denoted by C, and the size of the input is denoted by $H \times W$. The prediction output matrix is represented as $A \in \mathbb{R}^{C\times H\times W}$, which satisfies

$$\sum_{k=1}^{C} A_{k,i,j} = 1 \quad \forall i \in 1,\ldots,H, \quad j \in 1,\ldots,W \tag{1}$$

$$A_{k,i,j} \geq 0 \quad \forall k \in 1\ldots C, i \in 1,\ldots,H, \quad j \in 1,\ldots,W. \tag{2}$$

To measure the local consistency, several local prediction matrices are randomly cropped. The number of cropped regions is denoted by N_c, and the size of each local output matrix is denoted by $H_c \times W_c$. We now focus on one region, and the local matrix can be represented by $D \in \mathbb{R}^{C\times H_c\times W_c}$, which satisfies

$$\sum_{k=1}^{C} D_{k,i,j} = 1 \quad \forall i \in 1,\ldots,H_c, \quad j \in 1,\ldots,W_c \tag{3}$$

$$D_{k,i,j} >= 0 \quad \forall k \in 1\ldots C, i \in 1,\ldots,H_c, \quad j \in 1,\ldots,W_c. \tag{4}$$

It is clear that higher consistency means less uncertainty in the region. To measure the uncertainty, a common solution to strengthen the local prediction consistency is minimizing the Shannon entropy [28] of the local prediction matrices in the target domain [35]. The disadvantages of this method, which tends to reduce the diversity [37], can be relieved since the number of categories of objects in a small local region is relatively low in reality. The entropy of this local region can be calculated as follows:

$$H(D) = -\frac{1}{H_c \times W_c} \sum_{i=1}^{H_c}\sum_{j=1}^{W_c}\sum_{k=1}^{C} D_{k,i,j} \log D_{k,i,j}. \tag{5}$$

In order to better connect with the Frobenius norm that will be used, the local matrix can be reshaped to $C \times S$, where $S = H_c \times W_c$, so (3)–(5) can be modified as

$$\sum_{k=1}^{C} D_{k,i} = 1 \quad \forall i \in 1,\ldots,S \tag{6}$$

$$D_{k,i} \geq 0 \quad \forall k \in 1\ldots C, \quad i \in 1,\ldots,S \tag{7}$$

$$H(D) = -\frac{1}{S}\sum_{i=1}^{S}\sum_{k=1}^{C} D_{k,i}\mathrm{logD}_{k,i}. \tag{8}$$

We now introduce the Frobenius norm of the local matrix D and discuss the relationship between it and the local entropy. The Frobenius norm of the local matrix D is defined as

$$\|D\|_F = tr\left(D^T D\right) = \sqrt{\sum_{k=1}^{C}\sum\nolimits_{i=1}^{S} D_{ki}^2} = \sqrt{\sum\nolimits_{k=1}^{\min(C,S)} \sigma_i(D)} \tag{9}$$

where σ_i denotes the ith largest singular value.

Proposition 3.1: Given a local matrix D, $H(D)$ is bounded, which satisfies $0 \leq H(D) \leq \mathrm{logC}$.

Proof: On the one hand, it is clear that $H(D)$ is nonnegative because $\mathrm{logD}_i \leq 0$ and $Di \geq 0$. In addition, $H(D) = 0$ if and only if each D_i is a one-hot column vector. On the other hand, due to the concavity of function $f(x) = -\mathrm{xlogx}$, we have

$$\begin{aligned} H(D) &= -\frac{1}{S}\sum_{i=1}^{S}\sum_{k=1}^{C} D_{k,i}\mathrm{logD}_{k,i} \\ &= \frac{C}{S}\sum_{i=1}^{S}\left(-\frac{1}{C}\sum_{k=1}^{C} D_{k,i}\mathrm{logD}_{k,i}\right) \\ &\leq \frac{C}{S}\sum_{i=1}^{S}\left(-\frac{\sum_{k=1}^{C} D_{ki}}{C}\log\frac{\sum_{k=1}^{C} D_{ki}}{C}\right) \\ &=\leq \frac{C}{S}\sum_{i=1}^{S}\left(-\frac{1}{C}\log\frac{1}{C}\right) = \mathrm{logC} \end{aligned} \tag{10}$$

where $H(D)$ reaches logC when $D_{ij} = (1/C), \forall i, j$. □

Proposition 3.2: Given a local matrix D, $\|D\|_F$ is bounded, which satisfies $((S/C))^{1/2} \leq \|D\|_F \leq (S)^{1/2}$.

Proof: On the one hand

$$\begin{aligned} \|D\|_F &= \sqrt{\sum_{i=1}^{S}\sum_{k=1}^{C} D_{ki}^2} \\ &\leq \sqrt{\sum_{i=1}^{S}\left(\sum_{k=1}^{C} D_{ki}\right)^2} \\ &= \sqrt{\sum_{i=1}^{S} 1} = \sqrt{S} \end{aligned} \tag{11}$$

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where $\|D\|_F$ reaches the upperbound when each D_i is a one-hot column vector.

On the other hand, based on the Cauchy–Schwarz inequality [42], we have

$$\begin{aligned}\|D\|_F &= \sqrt{\sum_{i=1}^{S}\sum_{k=1}^{C} D_{ki}^2} \\ &\geq \sqrt{\sum_{i=1}^{S}\frac{\left(\sum_{k=1}^{C} D_{ki}\right)^2}{C}} \\ &= \sqrt{\sum_{i=1}^{S}\frac{1}{C}} = \sqrt{\frac{S}{C}}. \end{aligned} \tag{12}$$

where $\|D\|_F$ reaches the lower bound when each D_i is a probability vector with discrete uniform distribution. □

Proposition 3.3: The Frobenius norm $\|D\|_F$ is convex with respect to the matrix D.

Proof: Because every norm is convex [43], it is sufficient to prove that the Frobenius norm is, in fact, a norm.

It is trivial to verify that $\|D\|_F = 0$; then, $D = 0$ and $\|D\|_F = |t|\|D\|_F$ if t is a scalar. Therefore, by the definition of the nuclear norm, absolute homogeneity is correct.

It is only necessary to prove that the Frobenius norm satisfies the subadditivity or triangle inequality. It equals to prove that $\|A + B\|_F \leq \|A\|_F + \|B\|_F$, which can be calculated as follows:

$$\begin{aligned}|A+B\|_F^2 &= \sum_{i,j}\left(A_{ij}+B_{ij}\right)^2 \\ &= \sum_{i,j} A_{ij}^2 + \sum_{i,j} B_{ij}^2 + 2\sum_{i,j} A_{ij}B_{ij} \\ &\leq \sum_{i,j} A_{ij}^2 + \sum_{i,j} B_{ij}^2 + 2\sqrt{\left(\sum_{i,j} A_{ij}^2\right)\left(\sum_{i,j} B_{ij}^2\right)} \\ &= \left(\sqrt{\sum_{i,j} A_{ij}^2} + \sqrt{\sum_{i,j} B_{ij}^2}\right)^2 \\ &= (\|A\|_F + \|B\|_F)^2. \end{aligned} \tag{13}$$

□

Theorem 3.1: Given matrix D, $H(D)$ and $\|D\|_F$ have strict contrary monotonicity with respect to matrix D.

Proof: The proof can be found in Appendix A-A. □

Theorem 3.2: The minimum of $H(D)$ and the maximum of $\|D\|_F$ are achieved at the same solution of D.

Proof: According to Propositions 3.1 and 3.2, the minimum of $H(D)$ (i.e., 0) and the maximum of $\|D\|_F$ (i.e., $(S)^{1/2}$) are achieved at the same time when each D_i is a one-hot column vector. □

To sum up, the local consistency of prediction outputs can be enhanced not only by minimizing the Shannon entropy but also by maximizing the Frobenius norm of the local matrices. There are many advantages to making use of the Frobenius norm. First, it can be calculated faster than entropy and only requires little computation resources. Second, its scope can be kept consistent with the following global diversity metric, which promotes the balance between different loss functions. Third, it is well bounded, so it will not bring too much bias to the existing UDA training frameworks and keep the stability of the training process.

B. Measuring the Global Diversity With the Nuclear Norm

It is normal in randomly cropped N_c regions that some majority categories (e.g., agriculture) account for a large part, while other categories (e.g., barren) include less or even no pixels. In this case, the decision boundary of a model regularized with F-norm maximization or entropy minimization tends to approach the majority categories. The continuous adaptation to the majority categories reduces the prediction diversity, which is not what is wanted to see. To improve the overall accuracy, the aim is to maintain the prediction diversity by analyzing the N_c local matrices to model the global diversity.

To model global diversity, we start by looking at the occurrence probability of each category in each region. A global matrix $G \in \mathbb{R}^{N_c \times C}$ is defined, and we let the vector whose elements are the occurrence probability of the corresponding category in each region be the related row vector of the global matrix. In other words, G_{ij} is the appearance frequency of the jth category in the ith local prediction. To describe the meaning of the global matrix G and introduce the estimator based on the nuclear norm, we built three toy examples.

In the first example, it is assumed that N_c and C are 3. In addition, G is expressed as

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \tag{14}$$

In this case, when the three local prediction matrices are cropped from the whole result, it can be found that all the pixels are full of the 1th category. This means that the global diversity of the whole prediction is very poor because samples are only obtained from one class from the N_c randomly cropped regions. It can be found that the rank of G is 1, which suggests that a low rank may mean poor global diversity.

In the second example, it is assumed that N_c and C are 3. In addition, G is expressed as

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{15}$$

This case is the opposite situation of the first example. When the three local prediction matrices are cropped from the whole result, it can be found that the regions are filled with totally different categories. This means that the global diversity of the whole prediction is good because pixels are obtained from all classes from the N_c randomly cropped regions. It can be found that the rank of G is 3, which suggests that a high rank tends to mean great global diversity.

In the last example, it is assumed that N_c and C are 3. In addition, G is expressed as

$$G = \begin{bmatrix} 0.33 & 0.33 & 0.34 \\ 0.33 & 0.33 & 0.34 \\ 0.33 & 0.33 & 0.34 \end{bmatrix}. \tag{16}$$

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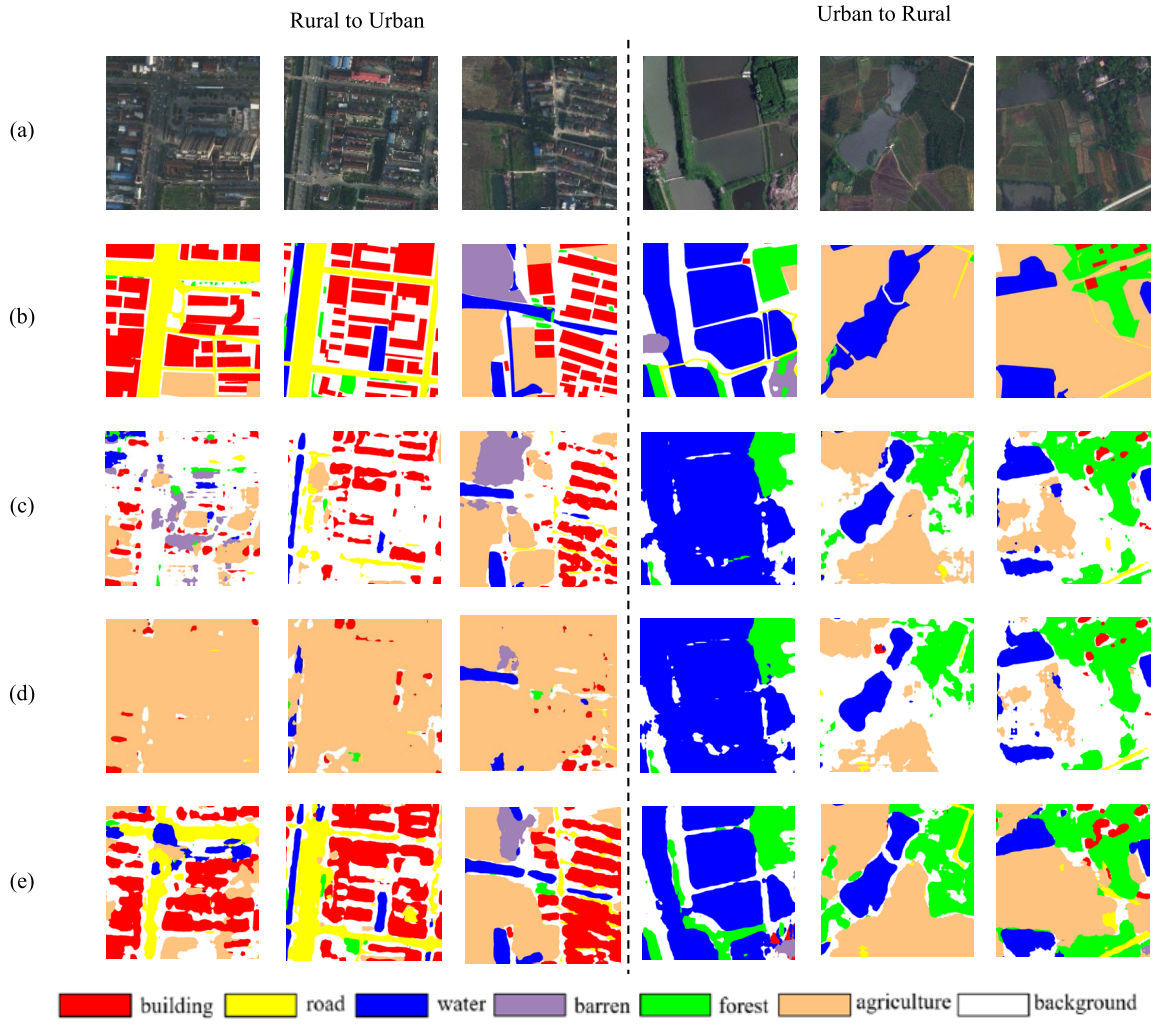


Fig. 2. Qualitative comparison between the proposed method and the other methods on the test set of the LoveDA dataset. (a) Image. (b) Ground truth. (c) CLAN. (d) CLAN+Ent. (e) CLAN+LCGDM.

This is also an extreme case, where each cropped region contains all C categories. There is no doubt that the diversity of this prediction is perfect because samples of all classes are obtained from the N_c randomly cropped regions. However, this violates the principle that the number of classes in a local region is always expected to be small, which is included in the meaning of good global diversity. It can be found that the rank of G is 1, which means that a low rank can also capture this deceptive "good" global diversity.

Based on the above findings, rank(G) can be an approximation to the global diversity. As the rank of G increases, the global diversity becomes better. However, the rank optimization problem has long been thought of as a difficult nonconvex problem. In order to solve this problem, we take advantage of the nuclear norm from matrix analysis [44] to replace the rank approximately. We now introduce the nuclear norm of the global matrix G and discuss its properties and advantages.

For the global matrix G, the nuclear norm $\|G\|_*$ is defined as the sum of the singular values of G, which is calculated as

$$G = \sum_{i=1}^{r} \sigma_i(G) \tag{17}$$

where σ_i denotes the ith largest singular value and $r = \min\{N_c, C\}$.

Proposition 3.4: Given a global matrix G, its nuclear norm $\|G\|_*$ is bounded by its Frobenius norm $\|G\|_F$, which satisfies $\|G\|_F \leq \|G\|_* \leq (r)^{1/2}\|G\|_F$.

Proof: On the one hand

$$\|G\|_*^2 = \left(\sum_{i=1}^{r} \sigma_i(G)\right)^2 \geq \sum_{i=1}^{r} \sigma_i^2(G) = \|G\|_F^2. \tag{18}$$

On the other hand, based on the Cauchy–Schwarz inequality [42], we have

$$\|G\|_* = \left(\sum_{i=1}^{r} \sigma_i(G)\right)^2 \leq r \sum_{i=1}^{r} \sigma_i^2(G) = r\|G\|_F^2. \tag{19}$$

By combining (18) and (19), we can easily obtain $\|G\|_F \leq \|G\|_* \leq (r)^{1/2}\|G\|_F$. □

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Proposition 3.5: The nuclear norm $\|G\|_*$ is convex with respect to the matrix G.

Proof: The proof is placed in Appendix A-B. □

Proposition 3.6: The convex envelope of rank(G) on the set $\{G \in \mathbb{R}^{N_c \times C} \leq 1\}$ is the nuclear norm $\|G\|_*$, where the convex envelope of a (possibly nonconvex) function $f : C \to R$ is defined as the largest convex function g such that $g(x) \leq f(x)$ for all $x \in C$.

Proof: The proof of this statement is given in [45]. This proof relies on von Neumann's trace inequality [44] and a basic result from convex analysis, which establishes that, under certain technical conditions, the biconjugate of a function is its convex envelope [46]. □

Theorem 3.3: Given a global matrix G, its nuclear norm $\|G\|_*$ is bounded and convex. In addition, minimizing $\|G\|_*$ is a good convex approximation to minimizing the rank of G.

Proof: First, according to Proposition 3.3, it is known that $\|G\|_F \leq \|G\|_* \leq (r)^{1/2}\|G\|_F$, where $r = \min\{N_c, C\}$. Similar to the proof of Proposition 3.2, we can obtain $((N_c/C))^{1/2} \leq \|G\|_F \leq (N_c)^{1/2}$. Combining these findings, we have

$$\sqrt{\frac{N_c}{C}} \leq \|G\|_* \leq \sqrt{r}\sqrt{N_c} \leq N_c. \tag{20}$$

Thus, the nuclear norm $\|G\|_*$ is bounded.

Second, it is directly known that $\|G\|_*$ is convex based on Proposition 3.4.

Third, we have $\|G\|_F \leq (N_c)^{1/2}$. Thus, the convex envelope of rank(G) becomes $\|G\|_*/(N_c)^{1/2}$, which is also proportional to $\|G\|_*$. As result, minimizing $\|G\|_*$ is a good convex approximation to minimizing the rank of G. □

From Theorem 3.3, we know that the global diversity of the land-cover prediction can be improved by maximizing the nuclear norm of the global matrix. We have proved many good properties of it. First, it is also bounded, so it will not bring too much bias as with the former metric. Second, it is convex rather than nonconvex as rank itself, so the optimization can be tackled easily. Third, it can be calculated quickly by using SVD, so there will be little extra computational consumption when adding this metric to the existing UDA training process.

C. Batch LCGDM Optimization for Domain Adaptation

In domain adaptation, there is a source domain $\mathcal{D}_S$ and a target domain $\mathcal{D}_T$. We denote them by $\mathcal{D}_S = \{(X_i^S, Y_i^S)\}_{i=1}^{N_S}$ and $\mathcal{D}_T = \{X_i^T\}_{i=1}^{N_T}$, which means that there are N_S labeled source samples in C categories and N_T unlabeled target samples in the same C categories. In $\mathcal{D}_S$, the source images and labels are denoted as $X_i^S \in \mathbb{R}^{3\times H\times W}$ and $Y_i^S \in \mathbb{R}^{C\times H\times W}$. Meanwhile, in $\mathcal{D}_T$, the target images are denoted as $X_i^T \in \mathbb{R}^{3\times H\times W}$.

For land-cover classification, the classification results are acquired by DNN S, i.e., $A_i = S(X_i)$. Typically, network S consists of a feature extraction network and a classifier. With a randomly sampled batch of B_S samples from the source domain and a batch of B_T samples from the target domain, it is possible to calculate the batch LCGDM and then optimize the neural network S.

1) Calculation and Properties of the Batch LCGDM: Previously, for convenience, we discussed the computational process of local consistency and global diversity when processing a single HRS image. However, randomly sampling a batch of HRS images and then optimizing the DNN are typically used in reality. The calculation of batch LCGDM is similar to above, which is shown in Algorithm 1, and its bound can be obtained as well.

Algorithm 1 Batch LCGDM Calculation

Input:

Neural network S, selected target data $X_{Bi}^T \in \mathbb{R}^{B_T\times 3\times H\times W}$ for the ith iteration, crop size (H_c, W_c), crop number N_c, weight of local consistency metric λ_{LCM},weight of global diversity metric λ_{GDM}, where $\lambda_{\text{GDM}} + \lambda_{\text{LCM}} = 1$.

Output:

Value of loss based on LCGDM $\mathcal{L}_{LCGDM}$.

$A \leftarrow S(X_{Bi}^T)$ $//A \in \mathbb{R}^{B_T\times C\times H\times W}$
$\mathcal{L}_{\text{LCM}} \leftarrow 0$
$G \leftarrow Null$
for $j = 1$ to N_c **do**
  $D^j \leftarrow Randomcrop(A, H_c, W_c)$
  $reshape(D^j) : \mathbb{R}^{B_T\times C\times H_c\times W_c} \to \mathbb{R}^{B_T\times C\times S}$
  $\mathcal{L}_{\text{LCM}} \leftarrow \mathcal{L}_{\text{LCM}} + \frac{1}{B_T}\sum_{i=1}^{B_T}\|D_i^j\|_F$
  $D^j \leftarrow sum(D, dim = 2)$ $//D^j \in \mathbb{R}^{B_T\times C}$
  $D^j \leftarrow softmax(D, dim = 1)$ $//D^j \in \mathbb{R}^{B_T\times C}$
  $G \leftarrow [G; D^j]$
end for
$//G \in \mathbb{R}^{B_T\times N_c\times C}$
$\mathcal{L}_{\text{LCM}} \leftarrow \frac{1}{N_c}\mathcal{L}_{\text{LCM}}$
$\mathcal{L}_{\text{GDM}} \leftarrow \frac{1}{B_T}\sum_{i=1}^{B_T}\|G_i\|_*$
$\mathcal{L}_{LCGDM} \leftarrow -(\lambda_{\text{LCM}}\mathcal{L}_{\text{LCM}} + \lambda_{\text{GDM}}\mathcal{L}_{\text{GDM}})$
return $\mathcal{L}_{LCGDM}$

Theorem 3.4: $\mathcal{L}_{\text{LCGDM}}$ is bounded.

Proof: First, because the Frobenius norm of each local matrix D_i^j in Algorithm 1 is bounded and satisfies $((S/C))^{1/2} \leq \|D_i^j\|_F \leq (S)^{1/2}$ according to Proposition 3.2, we prove that $\mathcal{L}_{\text{LCM}}$ is bounded directly

$$\begin{aligned}\mathcal{L}_{\text{LCM}} &= \frac{1}{N_c}\sum_{j=1}^{N_c}\frac{1}{B_T}\sum_{i=1}^{B_T}\|D_i\|_F \\ &\leq \frac{1}{N_c}\sum_{j=1}^{N_c}\frac{1}{B_T}\sum_{i=1}^{B_T}\sqrt{S} = \sqrt{S}\end{aligned} \tag{21}$$

$$\begin{aligned}\mathcal{L}_{\text{LCM}} &= \frac{1}{N_c}\sum_{j=1}^{N_c}\frac{1}{B_T}\sum_{i=1}^{B_T}\|D_i\|_F \\ &\geq \frac{1}{N_c}\sum_{j=1}^{N_c}\frac{1}{B_T}\sum_{i=1}^{B_T}\sqrt{\frac{S}{C}} = \sqrt{\frac{S}{C}}.\end{aligned} \tag{22}$$

Thus, $\mathcal{L}_{\text{LCM}}$ is bounded, and $((S/C))^{1/2} \leq \mathcal{L}_{\text{LCM}} \leq (S)^{1/2}$.

Second, because the nuclear norm of each global matrix G_i in Algorithm 1 is bounded and satisfies $((N_c/C))^{1/2} \leq \|G_i\|_* \leq (r)^{1/2}(N_c)^{1/2} \leq N_c$, according to Theorem 3.3, it can be examined that $\mathcal{L}_{\text{GDM}}$ is bounded

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directly

$$\mathcal{L}_{\mathrm{GDM}} = \frac{1}{B_T}\sum_{i=1}^{B_T}\|G_i\|_* \le \frac{1}{B_T}\sum_{i=1}^{B_T} N_c = N_c \tag{23}$$

$$\mathcal{L}_{\mathrm{GDM}} = \frac{1}{B_T}\sum_{i=1}^{B_T}\|G_i\|_* \ge \frac{1}{B_T}\sum_{i=1}^{B_T}\sqrt{\frac{N_c}{C}} = \sqrt{\frac{N_c}{C}}. \tag{24}$$

Thus, $((N_c/C))^{1/2} \le \mathcal{L}_{\mathrm{GDM}} \le N_c$.

Finally, we can prove this theorem by combining the two above findings

$$\begin{aligned}\mathcal{L}_{\mathrm{LCGDM}} &= -(\lambda_{\mathrm{LCM}}\mathcal{L}_{\mathrm{LCM}} + \lambda_{\mathrm{GDM}}\mathcal{L}_{\mathrm{GDM}}) \\ &\le -\left(\lambda_{\mathrm{LCM}}\sqrt{\frac{S}{C}} + \lambda_{\mathrm{GDM}}\sqrt{\frac{N_c}{C}}\right) \\ &\le -\sqrt{\frac{N_c}{C}}\end{aligned} \tag{25}$$

$$\begin{aligned}\mathcal{L}_{\mathrm{LCGDM}} &= -(\lambda_{\mathrm{LCM}}\mathcal{L}_{\mathrm{LCM}} + \lambda_{\mathrm{GDM}}\mathcal{L}_{\mathrm{GDM}}) \\ &\ge -\left(\lambda_{\mathrm{LCM}}\sqrt{S} + \lambda_{\mathrm{GDM}}N_c\right) \\ &\ge -N_c.\end{aligned} \tag{26}$$

Therefore, $\mathcal{L}_{\mathrm{LCGDM}}$ is bounded, and $-N_c \le \mathcal{L}_{\mathrm{LCGDM}} \le -((N_c/C))^{1/2}$. □

Benefiting from the computation-friendly and optimization-friendly proposed LCGDM, we can easily plugin it into the existing UDA training framework by calculating one more loss function as the subsequent subsections show.

2) Self-Training Methods With the LCGDM: Commonly, the UDA methods based on ST improve the performance on the target domain by generating and selecting pseudolabels in target data intermittently and then fine-tuning the model with these labels. We denote the pseudolabels of the ith batch of B_T samples from the target domain as $\hat{Y}^T_{Bi}$. The loss function in the ST methods' process then also generally consists of two parts, that is, $\mathcal{L}(X^S_{Bi}, Y^S_{Bi}, X^T_{B_i}, \hat{Y}^T_{Bi}) = \mathcal{L}_{\mathrm{seg}}(X^S_{Bi}, Y^S_{Bi}) + \lambda_{\mathrm{pse}}\mathcal{L}_{\mathrm{pse}}(X^T_{Bi}, \hat{Y}^T_{Bi})$, where $\mathcal{L}_{\mathrm{seg}}$ is the cross-entropy loss using the ground-truth annotations in the source domain, $\mathcal{L}_{\mathrm{pse}}$ is the cross-entropy loss between the target images and pseudolabels, and λ_{pse} is the weight used to balance the two losses.

The proposed $\mathcal{L}_{\mathrm{LCGDM}}$ can be integrated into the optimization process of the existing UDA algorithms based on ST as well, which is shown in Algorithm 2.

3) Adversarial Training Methods With the LCGDM: Typically, UDA algorithms based on AT consist of two modules: a segmentation network S and a discriminator Dis. These algorithms minimize several adversarial loss functions to align the source and target data distributions and make use of a discriminator to distinguish the domain-invariant features captured by the feature extraction network S. The adaptation tasks are always formulated to contain two loss functions from both modules

$$\begin{aligned}&\mathcal{L}\left(X^S_{Bi}, Y^S_{Bi}, X^T_{B_i}\right) \\ &\quad = \mathcal{L}_{\mathrm{seg}}\left(X^S_{Bi}, Y^S_{Bi}\right) + \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}}\left(X^S_{Bi}X^T_{B_i}\right)\end{aligned} \tag{27}$$

where $\mathcal{L}_{\mathrm{seg}}$ is the cross-entropy loss using the ground-truth annotations in the source domain and $\mathcal{L}_{\mathrm{adv}}$ is the adversarial loss that adapts the predicted segmentation of the target images to the distribution of the source predictions. λ_{adv} is the weight used to balance the two losses.

Our loss function $\mathcal{L}_{\mathrm{LCGDM}}$ can be integrated into the optimization process of the existing UDA algorithms based on AT, which is displayed in Algorithm 3. We note that many existing AT training processes do not have the warm-up stage. To keep the consistency between Algorithms 2 and 3, we introduce the warm-up iterations to AT training as well. Experiment results in the later section support this change, that is, a suitable warm-up can improve the performance of AT training.

Algorithm 2 ST Methods With LCGDM

Input:
Neural network S, selected batch of source data (X^S_{Bi}, Y^S_{Bi}) and target data X^T_{Bi} for the ith iteration, crop size (H_c, W_c), crop number N_c, weight λ_{LCM} of $\mathcal{L}_{\mathrm{LCM}}$, weight λ_{GDM} of $\mathcal{L}_{\mathrm{GDM}}$, weight λ_{LCGDM} of $\mathcal{L}_{\mathrm{LCGDM}}$, W warm-up iterations, total M iterations, and K.

Output:
Network weights of S.

for $i = 1$ to M **do**
  if $i \le W$ **then**
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{seg}}$
  else
    if $i\%K = 0$ **then**
      $Y^T \leftarrow$ generate pseudo-labels
    end if
    $\mathcal{L}_{LCGDM} \leftarrow$ Algorithm1(S, X^T_{Bi}, H_c, W_c, N_c, λ_{GDM}, λ_{GDM})
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{pse}}\mathcal{L}_{\mathrm{pse}} + \lambda_{LCGDM}\mathcal{L}_{LCGDM}$
  end if
  Update S
end for
return S

Algorithm 3 AT Methods With LCGDM

Input:
Neural network S, discriminator Dis, selected batch of source data (X^S_{Bi}, Y^S_{Bi}) and target data X^T_{Bi} for the ith iteration, crop size (H_c, W_c), crop number N_c, weight λ_{LCM} of $\mathcal{L}_{\mathrm{LCM}}$, weight λ_{GDM} of $\mathcal{L}_{\mathrm{GDM}}$, weight λ_{LCGDM} of $\mathcal{L}_{LCGDM}$, W warm-up iterations, total M iterations.

Output:
Network weights of S.

for $i = 1$ to M **do**
  if $i \le W$ **then**
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}}$
  else
    $\mathcal{L}_{LCGDM} \leftarrow$ Algorithm1(S, X^T_{Bi}, H_c, W_c, N_c, λ_{GDM}, λ_{GDM})
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}} + \lambda_{LCGDM}\mathcal{L}_{LCGDM}$
  end if
  Update S
end for
return S

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IV. Experiments

We applied the proposed LCGDM to two UDA frameworks, i.e., AT (AdaptSeg [15], FADA [16], and CLAN [17]), ST (PyCDA [19], CBST [20], and IAST [21]), and RS [22], [34]. The experiments with the two approaches were accomplished on the LoveDA [30] dataset. We conducted an ablation study to prove the effectiveness of proposed LCM and GDM. We also conducted sensitivity analyses for hyperparameters, i.e., the crop number N_c, the crop size (H_c, W_c), the number of warm-up iterations W, and the weight for LCGDM λ_{LCGDM}. The adequate experiments and visual results prove the validity and flexibility of the proposed LCGDM with existing UDA frameworks.

A. Dataset Description

The LoveDA dataset [30] contains 5987 HSR (1024 × 1024) images with 166 768 annotated objects from three different cities. Compared to the existing datasets, the LoveDA dataset encompasses two domains (urban and rural), which also brings considerable challenges due to the multiscale objects, complex background samples, and inconsistent class distributions. This dataset requires a DNN model with better generalization capability for the different urban and rural scenes. For the rural-to-urban task, there are 1366 source images (rural) and 820 target images (urban). As for the urban-to-rural task, there are 1156 source images (urban) and 976 target images (rural).

B. Experimental Settings

1) Experimental Environment: All the experiments are performed using a single TITAN X GPU with 12-GB memory. The implementation of the framework was based on the open-source Pytorch library [47]. In this article, we report the average results of three times experiments.

2) Implementation Details: The intersection over union (IoU) is used to measure the semantic segmentation quality. With respect to the IoU for each class, the mIoU represents the mean of the IoUs over all the classes.

In terms of the neural network architecture, all the methods adopted the same feature extractor and discriminator (for methods based on AT), following the common implementation [15], [16], [17]. Specifically, DeepLabv2 [48] with ResNet50 [12] was utilized as the extractor, and the discriminator was composed of five convolutional layers with 4 × 4 kernel with channel numbers {64, 128, 256, 512, 1} and stride 2. Each convolutional layer was followed by a leaky rectified linear unit (Leaky-ReLU) parameterized by 0.2, except for the last layer. The backbones used in all the networks were pretrained on ImageNet [49].

The total number of training iterations M was set to 20k, with a batch size of 8. The number of warm-up iterations was set to 6k. For the data augmentation, 512 × 512 patches were randomly cropped from the raw inputs, with random mirroring and rotation. During the training, we use the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 10^{-4} to update the feature extractor.

In the case of AT, the learning rates of the main model and discriminator were set to 5×10^{-3} and 10^{-4}, respectively. The Adam optimizer was used for the discriminator with a momentum of 0.9 and 0.99. In the case of ST, the pseudolabels were generated every 1k iteration.

C. Comparison With the Existing Methods

Based on the existing advanced AT and ST frameworks, we integrated the proposed LCGDM into their training processes. The results are displayed in Tables I and II. It can be found that, by adding the LCGDM as an auxiliary loss function, both the categorywise performance and overall performance can be clearly improved. It is worth noting that the hyperparameters are not tuned exactly, and we just kept $W_c = H_c = 4$ and $N_c = 7$. In addition, when the basic algorithms were based on ST, we picked 0.5 as the reweighting parameter λ_{pse}. We chose 0.01, 0.01, and 0.1 as λ_{pse} for AdaptSeg, FADA, and CLAN, respectively. Notably, when applying domain adaptation from rural to urban and fixing CLAN as the basic framework, making use of the LCGDM can improve the mIOU from 33.78% to 39.35%. Furthermore, when applying domain adaptation from urban to rural and fixing IAST as the basic framework, making use of the LCGDM can improve the mIOU from 36.34% to 40.86%.

Results in Tables I and II can guide us to select training framework when we meet HRS images. We note that, though there exist inconsistent performance improvements in urban to rural and rural to urban settings, for example, the method (CBST) that achieves the best performance in the rural to urban setting fails to take the crown in the urban to rural setting, methods based on ST always perform better than that based on AT. This phenomenon suggests the superiority of ST in the RS field. We think that the reason is that AT needs more samples to train the extra discriminator relative to ST. As the HRS images are much more complicated than natural images, this hunger for data is strengthened.

We also compared the LCGDM with the algorithm proposed in [26], which we denote as "Ent." We found that the improvement effect of Ent on the domain adaptation performance is not very stable. For instance, when we chose FADA as the baseline model, adding Ent results in a decrease in the mIOU. Though Ent enhances the land-cover mapping, its effect is not as good as that of the LCGDM. For example, when we applied CBST to the rural-to-urban task, adding Ent can improve the mIOU from 41.92% to 43.22%, while using the LCGDM can improve it to 44.99%.

We can explain this phenomenon to some extent based on both theory and the results. First, since Ent [26] aims to minimize the mean entropy of the prediction probability of each pixel, models trained with this technology tend to divide one pixel into the major category, which counts a lot in the whole dataset. Besides, the diversity of the prediction is likely to be reduced [37]. Second, our results in Table II give us some evidence. It could be found that, in urban to rural settings, methods with Ent always have more prediction accuracy in the agriculture category but have lower performance in the mean sense (mIOU). This suggests that methods with Ent tend

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TABLE I
SEMANTIC SEGMENTATION RESULTS OBTAINED ON THE LOVEDA TEST SET (RURAL TO URBAN)

Domain	Method	Type	IoU (%)							mIoU(%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
Rural to Urban	Oracle	-	48.18	52.14	56.81	85.72	12.34	36.70	35.66	46.79
	Source only	-	43.3	25.63	12.70	76.22	12.52	23.34	25.14	31.27
	AdaptSeg [15]	AT	41.32	26.42	15.23	80.83	15.87	34.94	20.93	33.65
	AdaptSeg+Ent [26]	AT	43.65	27.44	15.02	80.70	15.89	35.10	23.00	34.40
	AdaptSeg+LCGDM	AT	42.54	33.73	39.25	74.65	15.01	39.24	23.07	38.21 (+4.56)
	FADA [16]	AT	37.94	19.09	6.90	80.46	14.87	32.33	18.57	30.02
	FADA+Ent [26]	AT	37.57	10.07	2.35	78.58	13.52	37.34	19.75	28.45
	FADA+LCGDM	AT	32.48	31.91	25.16	69.66	11.40	40.50	18.47	32.80 (+2.78)
	CLAN [17]	AT	43.44	23.48	20.48	77.37	13.62	34.62	23.46	33.78
	CLAN+Ent [26]	AT	42.96	12.28	6.51	80.88	14.49	34.97	22.21	30.61
	CLAN+LCGDM	AT	37.10	43.05	40.04	67.15	18.22	42.24	27.67	39.35 (+5.57)
	PyCDA [19]	ST	22.61	22.54	34.39	78.07	12.09	37.07	13.11	31.41
	PyCDA+Ent [26]	ST	29.39	21.44	32.52	80.40	13.81	40.08	15.54	33.31
	PyCDA+LCGDM	ST	21.12	28.32	39.51	82.76	13.84	39.95	15.59	34.45 (+3.04)
	CBST [20]	ST	44.78	42.76	38.33	75.96	14.23	39.16	38.19	41.92
	CBST+Ent [26]	ST	**49.56**	43.17	46.22	**86.91**	18.69	20.84	37.15	43.22
	CBST+LCGDM	ST	47.09	**49.91**	48.16	84.23	18.05	32.06	35.49	**44.99** (+3.07)
	IAST [21]	ST	48.77	32.40	39.07	83.17	19.28	16.63	31.92	38.75
	IAST+Ent [26]	ST	47.56	35.39	18.50	84.34	19.72	22.11	34.52	37.45
	IAST+LCGDM	ST	48.26	24.81	**49.28**	85.85	19.53	39.33	27.34	42.06 (+3.31)
	DCA [34]	RS	38.28	32.31	29.56	74.70	9.09	**40.57**	18.48	34.71
	DCA+Ent [26]	RS	39.94	45.52	33.61	78.24	10.50	37.68	19.28	37.82
	DCA+LCGDM	RS	39.52	37.41	35.70	78.36	13.24	39.50	24.42	38.31 (+3.6)
	LoveCS [22]	RS	46.63	42.58	45.89	83.57	16.33	13.29	27.66	39.42
	LoveCS+Ent [26]	RS	48.01	26.23	36.81	83.83	**19.84**	27.24	**41.30**	40.47
	LoveCS+LCGDM	RS	42.79	44.76	42.98	77.74	13.10	28.05	37.71	41.02 (+1.6)

to choose agriculture as their result, which accounts for the largest proportion in the rural part of the LoveDA dataset.

LCGDM avoids this bias better. On the one hand, the shift to the major category can be relieved since the number of categories of objects in a small local region is relatively low naturally. On the other hand, the global diversity metric is proposed to boost the diversity of the prediction. These support the superiority of our method.

D. Ablation Study

The proposed method is inspired by the local consistency and global diversity of HRS images. However, up until now, it has not been known how these factors affect semantic segmentation performance. We, therefore, conducted an ablation study to analyze the effects of these factors. The results are listed in Tables III and IV. We picked CBST as the baseline model for the rural-to-urban adaptation and CLAN for the urban-to-rural task because these methods are effective examples of ST and AT frameworks, respectively. In Tables III and IV, using the LCM denoted by $LD = 1$ and using the GDM denoted by $GD = 1$. When $LD = 1$ and $GD = 1$ at the same time, it implies that LCGDM is used. The experiments in both rural-to-urban and urban-to-rural tasks show that using the LCGDM is better than applying either the LCM or GDM alone. We think that the reason for this phenomenon is that consistency and diversity are conflicting to some extent. Thus, enhancing the consistency/diversity alone can make the diversity/consistency worse. However, the LCGDM considers both perspectives and can find their Pareto-optimal results, which results in a better land-cover classification performance.

E. Hyperparameter Analysis

To explore the sensitivity of the hyperparameters, i.e., the warm-up iterations W, the crop size (H_c, W_c), the crop number N_c, and weights for the proposed losses λ_{LCGDM}, we conducted a series of experiments on the LoveDA dataset as follows.

As the benefits of warm-up operation have been proved in many ST methods [19], [20], [21], we consider analyzing the results by introducing warm-up iterations W to the AT training process. We chose CLAN as the basic framework and adjusted W from 0 to M, where M was fixed as 20k.

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TABLE II
SEMANTIC SEGMENTATION RESULTS OBTAINED ON THE LOVEDA TEST SET (URBAN TO RURAL)

Domain	Method	Type	IoU (%)							mIoU(%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
Urban to Rural	Oracle	-	37.07	53.72	43.99	64.89	10.18	44.73	61.11	45.10
	Source only	-	26.30	37.12	29.63	51.10	9.61	35.08	46.84	33.67
	AdaptSeg [15]	AT	23.60	41.54	25.90	46.77	12.39	33.42	36.14	31.40
	AdaptSeg + Ent [26]	AT	26.44	45.19	30.40	47.54	12.33	35.44	37.62	33.57
	AdaptSeg + LCGDM	AT	24.95	43.56	26.96	55.89	16.83	35.24	33.87	33.90 (+2.50)
	FADA [16]	AT	25.28	31.04	28.09	45.18	19.98	36.23	24.25	30.01
	FADA+Ent [26]	AT	24.00	31.65	27.51	46.84	20.94	35.70	19.06	29.39
	FADA+LCGDM	AT	32.48	31.91	25.16	69.66	11.40	40.50	18.47	32.80 (+2.79)
	CLAN [17]	AT	24.21	44.61	27.6	46.83	15.78	37.34	24.65	31.57
	CLAN+Ent [26]	AT	22.54	40.97	26.26	48.95	16.61	36.80	20.95	30.44
	CLAN+LCGDM	AT	24.80	37.78	26.57	56.99	22.63	37.3	39.00	35.01 (+3.44)
	PyCDA [19]	ST	18.17	49.85	19.37	52.95	30.72	35.72	32.85	34.23
	PyCDA+Ent [26]	ST	12.98	39.64	17.83	56.85	22.31	36.91	36.19	31.82
	PyCDA+LCGDM	ST	16.98	44.96	18.14	59.70	28.66	36.03	36.46	34.42 (+0.19)
	CBST [20]	ST	27.88	37.53	35.90	43.26	31.01	34.07	53.67	37.62
	CBST+Ent [26]	ST	23.81	46.13	26.90	66.44	17.39	35.79	**64.90**	40.19
	CBST+LCGDM	ST	**29.13**	35.71	32.90	54.75	**41.54**	34.70	56.56	40.76 (+3.14)
	IAST [21]	ST	23.66	**53.95**	20.74	60.68	8.77	32.18	62.30	36.34
	IAST+Ent [26]	ST	26.40	51.88	22.69	65.37	7.66	32.30	61.78	38.30
	IAST+LCGDM	ST	24.88	52.50	26.15	**66.79**	24.16	33.04	58.49	**40.86** (+4.52)
	DCA [34]	RS	26.42	41.88	33.02	55.44	29.78	22.26	39.21	31.60
	DCA+Ent [26]	RS	23.47	48.72	36.28	48.56	5.63	23.77	10.45	28.12
	DCA+LCGDM	RS	25.98	45.91	34.60	58.45	0.26	26.31	34.81	32.33 (+0.73)
	LoveCS [22]	RS	27.34	49.89	34.87	53.11	11.53	36.05	46.03	36.98
	LoveCS+Ent [26]	RS	24.59	45.90	40.19	59.59	3.59	29.86	45.01	35.53
	LoveCS+LCGDM	RS	28.53	35.51	33.92	62.52	11.33	**38.23**	50.67	37.25 (+0.27)

TABLE III
ABLATION STUDY RESULTS OBTAINED ON THE LOVEDA TEST SET (RURAL TO URBAN)

Domain	Method	LD	GD	IoU (%)							mIoU(%)
				Background	Building	Road	Water	Barren	Forest	Agriculture	
Rural to Urban	CBST	1	0	**48.09**	45.43	42.81	83.01	17.18	25.53	23.77	40.83
		0	1	46.30	45.65	37.71	83.56	10.62	33.54	43.29	42.95
		1	1	47.09	**49.91**	**48.16**	**84.23**	18.05	32.06	35.49	**44.99**
	CLAN	1	0	44.32	21.49	21.84	82.22	9.31	36.30	24.01	34.21
		0	1	42.61	13.10	16.25	81.91	15.81	23.25	19.27	30.31
		1	1	37.10	43.05	40.04	67.15	**18.22**	**42.24**	27.67	39.35

The results are displayed in Table V. The experiments in both urban-to-rural and rural-to-urban situations show that training with the LCGDM can always perform better than training without it, regardless of how W is set. In addition, W should be set moderately. When the W is too small, the prediction performance of the baseline model is not good enough, so applying regularization to it may even bring side effects. Furthermore, when W is too large, it is too late to boost the performance by using LCGDM. To find an optimal W in the practical experiments, one can run the binary search in the range from 0 to M, which can output a good-performing W in $O(\log M)$ tries.

In terms of the crop size (H_c, W_c) and the crop number N_c, we let $S_c = H_c = W_c$, for simplicity. We analyzed the performance of these hyperparameters by setting S_c and N_c in a range from 4 to 14. We chose CBST as the baseline model

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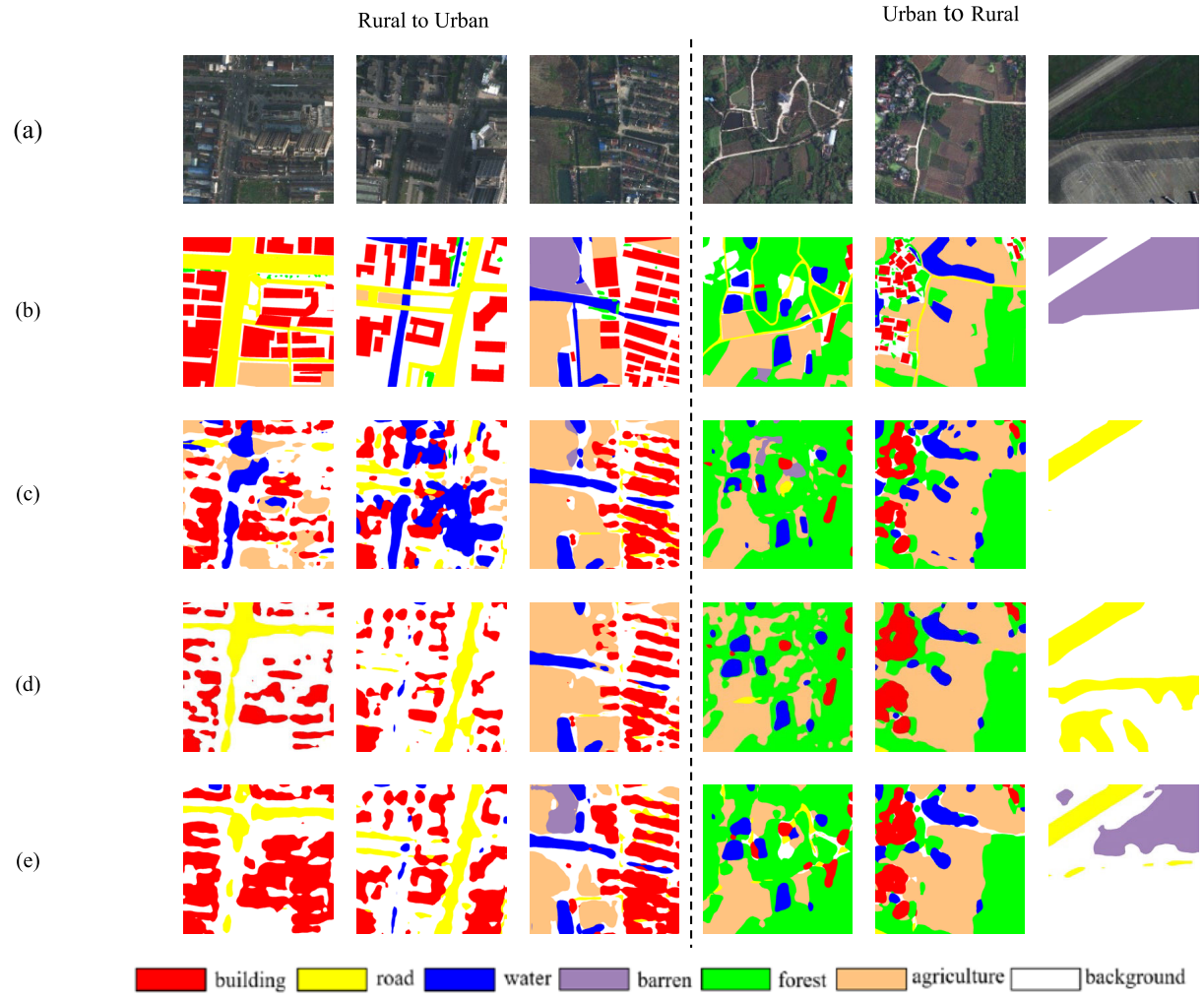


Fig. 3. Qualitative comparison between the proposed method and the other methods on the test set of the LoveDA dataset. (a) Image. (b) Ground truth. (c) CBST. (d) CBST+Ent. (e) CBST+LCGDM.

TABLE IV
ABLATION STUDY RESULTS OBTAINED ON THE LOVEDA TEST SET (URBAN TO RURAL)

Domain	Method	LD	GD	IoU (%)							mIoU(%)
				Background	Building	Road	Water	Barren	Forest	Agriculture	
Urban to Rural	CBST	1	0	23.44	41.26	**35.89**	**61.85**	6.50	30.22	51.93	35.88
		0	1	22.19	39.47	30.80	56.57	19.52	37.52	53.45	37.08
		1	1	**29.13**	35.71	32.90	54.75	**41.54**	34.70	**56.56**	**40.76**
	CLAN	1	0	22.86	40.29	27.64	46.49	14.14	**38.38**	22.08	30.27
		0	1	23.66	**41.35**	23.63	53.52	15.69	38.27	29.80	32.27
		1	1	24.80	37.78	26.57	56.99	22.63	37.30	39.00	35.01

and applied it to the rural-to-urban task with the LoveDA dataset. The results are listed in Table VI. It can be found that using the LCGDM with different values of S_c and N_c always outperforms the baseline model. However, to obtain the optimal performance of the LCGDM, N_c and S_c should be picked carefully because they are related to the natural properties of the HRS image dataset. Furthermore, the optimal values of N_c and S_c will differ for different datasets.

To analyze the weights for proposed losses λ_{LCGDM}, we chose CBST and CLAN as the basic frameworks and adjusted λ_{LCGDM} from 10^{-2} to 0.25 and from 10^{-3} to 10^{-1}, respectively. The results are displayed in Table VII. The experiments in the rural-to-urban setting show that λ_{LCGDM} should be fine-tuned carefully, especially when the basic framework is AT. Techniques such as hyperparameter optimization could be used to find an optimal weight for LCGDM.

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TABLE V
HYPERPARAMETER ANALYSIS FOR W ON THE LOVEDA TEST SET

Method	Domain	W	IoU (%)							mIoU(%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
CLAN	Urban to Rural	0	23.73	45.59	19.74	54.77	17.06	34.24	30.86	32.28
		0.1M	25.01	43.07	24.59	56.82	24.26	35.42	38.29	35.35
		0.2M	24.27	43.29	23.55	55.20	18.97	39.03	37.42	34.53
		0.3M	24.80	37.78	26.57	56.99	22.63	37.3	39.00	35.01
		0.4M	22.67	34.27	26.81	55.18	22.65	36.66	30.00	32.61
		0.5M	24.20	42.90	25.00	54.24	20.26	36.16	25.86	32.66
	Rural to Urban	0	38.13	44.17	42.68	55.11	13.90	38.97	23.98	36.71
		0.1M	34.04	46.01	46.41	61.34	14.52	39.51	23.81	37.95
		0.2M	30.54	43.17	47.68	69.39	12.43	38.12	19.81	37.31
		0.3M	37.10	43.05	40.04	67.15	18.22	42.24	27.67	39.35
		0.4M	38.45	40.14	46.76	48.10	13.67	29.64	15.90	33.24
		0.5M	31.63	36.11	46.99	38.91	14.47	35.80	21.86	32.25

TABLE VI
HYPERPARAMETER ANALYSIS FOR N_c AND S_c OBTAINED ON THE LOVEDA TEST SET (RURAL TO URBAN)

mIoU (%) N_c \ S_c	4	7	10	14
4	40.77	41.29	41.53	42.36
7	**44.99**	44.02	41.14	41.01
10	43.61	41.91	42.66	42.39
14	41.96	41.745	41.15	41.66
Basline	41.92			

F. Visualization Results

The proposed LCGDM was compared with other methods based on both AT and ST frameworks. Qualitative comparisons were also conducted in both the rural-to-urban and urban-to-rural settings. The visualization results are displayed in Figs. 2 and 3. It can be seen that the proposed LCGDM can produce better land-cover predictions than other methods and is more stable than using Ent [26] as the auxiliary loss function. Specifically, in the rural-to-urban experiments, the LCGDM predicts more complete results for the roads and buildings. Meanwhile, in the urban-to-rural experiments, the land-cover results for the water, barren, and agriculture classes are superior. The visualization results confirm the validity of the proposed LCGDM.

In order to verify the effects of the proposed LCGDM, several representative images were selected from the urban domain and then processed with models trained with different processes. The high-level semantic features were reduced to a 2-D space with t-distributed stochastic neighbor embedding (t-SNE [50]). The results can be seen in Fig. 4. The features of the baselines are easily confused. In contrast, by adding the proposed LCGDM, they achieve a better feature transformation, which is clear when the basic framework is CLAN.

V. DISCUSSION

As a plug-in module, the LCGDM works well on many existing UDA methods, which gives us some inspiration. In this section, we discuss how to design a good UDA method for practical RS tasks and the selection of the UDA architecture when faces with new tasks.

LCGDM gives us a new way to improve the UDA performance for RS tasks, that is, rethinking the natural properties of HRS images in different tasks. In this article, we only consider local consistency and global diversity of HRS images. However, more advanced topics can be considered, for example, the shape of different objects. We know that the buildings are always in a square shape, the roads are always in a line shape, the agriculture and lakes are always over a large area, and so on. This idea can also be applied to other tasks by proposing some task-specified prior properties metric.

In terms of the UDA training process and architecture, based on our experiments, we recommend the ST algorithm, which is more stable than AT. A promising way is to use LCGDN based on the emerging state-of-the-art ST methods. As for the neural architecture, one can also consider other neural networks, such as the graph neural network [51] and the transformer [52]. For instance, when the task needs modeling for long-distance dependence, we can choose a transformer as our backbone.

VI. CONCLUSION

In this article, we have focused on the very nature of HRS images and proposed a novel LCGDM. We use the Frobenius norm of the local prediction matrix and the nuclear norm of the global diversity matrix to, respectively, measure the local consistency and global diversity of land-cover classification results. The proposed LCGDM can be computed quickly and can be flexibly integrated into the existing UDA frameworks by reweighting the loss function. Extensive experiments conducted on the LoveDA dataset demonstrated the effectiveness of the proposed LCGDM.

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TABLE VII
HYPERPARAMETER ANALYSIS FOR λ_{LCGDM} ON THE LOVEDA TEST SET

Domain	Method	λ_{LCGDM}	IoU (%)							mIoU(%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
Rural to Urban	CBST	0.25	47.09	49.91	48.16	84.23	18.05	32.06	35.49	44.99
		0.05	46.31	38.06	44.15	83.71	11.89	27.83	26.08	39.72
		0.01	45.77	44.35	40.39	81.22	9.28	41.93	39.31	43.18
	CLAN	0.1	37.10	43.05	40.04	67.15	18.22	42.24	27.67	39.35
		0.01	40.85	28.97	18.79	75.14	12.86	40.23	25.08	34.56
		0.001	44.00	22.51	20.72	81.19	8.54	36.83	24.58	34.05

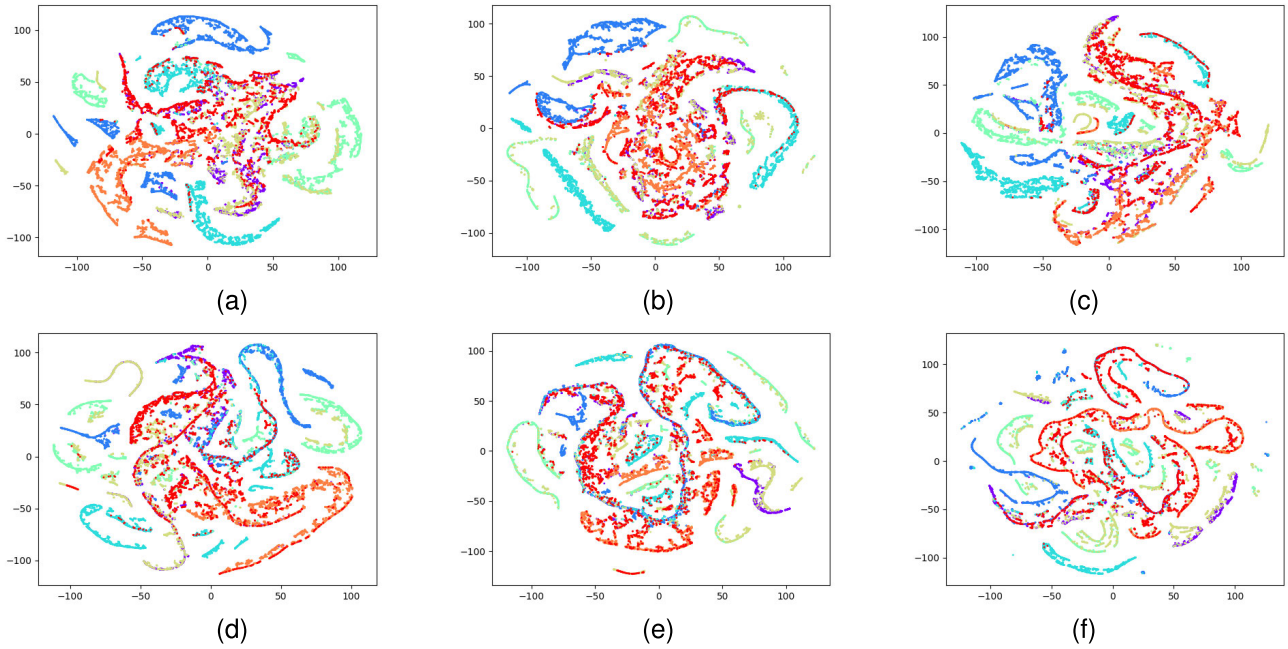


Fig. 4. Effects of the proposed LCGDM shown through t-SNE [50] visualization of the learned representations in the rural to urban setting. (a) CBST. (b) CBST+Ent. (c) CBST+LCGDM. (d) CLAN. (e) CLAN+Ent. (f) CLAN+LCGDM.

In the future, we will pay more attention to the properties of the HRS images and develop a more accurate metric to boost the UDA methods. Moreover, we will apply these findings to other tasks in the RS field, such as object detection, change detection, and instance segmentation.

APPENDIX A
PROOFS

A. Proof of Theorem 3.1

Proof: To prove that $H(D)$ and $\|D\|_F$ have strict contrary monotonicity, we can prove that $H(D)$ and $\|D\|_F^2$ have the same property, and the latter is denoted as $f(D)$. We start by deriving them and obtain that $(\partial f(D)/\partial D) = 2D$ and $(\partial H(D)/\partial D) = -\text{logD} - E$, where $(\text{logD})_{ij} = \log(D_{ij})$ and $E_{ij} = 1$.

To prove the theorem, all we need to prove is

$$(f(D+\Delta) - f(D))(H(D+\Delta) - H(D)) \le 0 \tag{28}$$

where $\sum_{i=1}^{C} D_{ij} = 1, \sum_{i=1}^{C} \Delta_{ij} = 0$, and $\|\Delta\|_F \to 0$. This can be simplified as follows:

$$\begin{aligned}
&(f(D+\Delta) - f(D))(H(D+\Delta) - H(D)) \\
&= \left(\sum_{i,j} \frac{\partial f(D)}{\partial D_{ij}} \Delta_{ij}\right)\left(\sum_{i,j} \frac{\partial H(D)}{\partial D_{ij}} \Delta_{ij}\right) \\
&= \left(\sum_{i,j} 2D_{ij}\Delta_{ij}\right)\left(\sum_{i,j} -(\text{logD}_{ij} + 1)\Delta_{ij}\right) \\
&= -2\left(\sum_{i,j} D_{ij}\Delta_{ij}\right)\left(\sum_{i,j} \text{logD}_{ij}\Delta_{ij}\right).
\end{aligned} \tag{29}$$

Thus, what we need to prove now is

$$\left(\sum_{i,j} D_{ij}\Delta_{ij}\right)\left(\sum_{i,j} \text{logD}_{ij}\Delta_{ij}\right) \ge 0. \tag{30}$$

The above formulas can be simplified as follows:

$$\begin{aligned}
&\sum_{i,j} D_{ij}\Delta_{ij} \\
&= \sum_{j=1}^{S}\sum_{i=1}^{C} D_{ij}\Delta_{ij}
\end{aligned}$$

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$$
\begin{aligned}
&= \sum_{j=1}^{S}\sum_{i=2}^{C}\left(D_{ij} - D_{1j}\right)\Delta_{ij} \\
&= \sum_{i\neq 1,\, j}\left(D_{ij} - D_{1j}\right)\Delta_{ij} \qquad (31)\\
&\sum_{i,j}\mathrm{logD}_{ij}\Delta_{ij} \\
&= \sum_{j=1}^{S}\sum_{i=1}^{C}\mathrm{logD}_{ij}\Delta_{ij} \\
&= \sum_{j=1}^{S}\sum_{i=2}^{C}\left(\mathrm{logD}_{ij} - \mathrm{logD}_{1j}\right)\Delta_{ij} \\
&= \sum_{i\neq 1,\, j}\left(\mathrm{logD}_{ij} - \mathrm{logD}_{1j}\right)\Delta_{ij} \\
&= \sum_{i\neq 1,\, j}\log\left(1 + \frac{D_{ij} - D_{1j}}{D_{1j}}\right)^{\Delta_{ij}} \\
&= \log\prod_{i\neq 1,\, j}\left(1 + \frac{D_{ij} - D_{1j}}{D_{1j}}\right)^{\Delta_{ij}} \\
&= \log\prod_{i\neq 1,\, j} e^{\left(1+\frac{D_{ij}-D_{1j}}{D_{1j}}\right)\Delta_{ij}} \\
&= \log\prod_{i\neq 1,\, j}\left(1 + \left(1 + \frac{D_{ij} - D_{1j}}{D_{1j}}\right)\Delta_{ij} + o(\|\Delta\|_F)\right) \\
&= \log\left(1 + \sum_{i\neq 1,\, j}\left(1 + \frac{D_{ij} - D_{1j}}{D_{1j}}\right)\Delta_{ij} + o(\|\Delta\|_F)\right) \\
&= \log\left(1 + \sum_{i\neq 1,\, j}\frac{D_{ij} - D_{1j}}{D_{1j}}\Delta_{ij} + o(\|\Delta\|_F)\right) \\
&\to \log\left(1 + \sum_{i\neq 1,\, j}\frac{D_{ij} - D_{1j}}{D_{1j}}\Delta_{ij}\right). \qquad (32)
\end{aligned}
$$

By combining the results of (31) and (32), we can know that inequality (30) is correct. □

B. Proof of Theorem 3.5

Proof: Because every norm is convex [43], it is sufficient to prove that the nuclear norm is, in fact, a norm.

First, it is trivial to verify that, if $\|G\|_* = 0$, then $G = 0$. Second, to prove the absolute homogeneity (i.e., $\|tG\|_* = |t|\|G\|_*$ if t is a scalar), we then have

$$tG = t\left(V\Sigma W^T\right) = (\mathrm{sign}(t)V)(|t|\Sigma)W^T. \qquad (33)$$

Therefore, by the definition of the nuclear norm, absolute homogeneity is correct.

Third, to prove that the nuclear norm satisfies the subadditivity or triangle inequality, we need to prove that $\|A + B\|_* \leq \|A\|_* + \|B\|_*$. To do this, it is first necessary to prove that the nuclear norm is dual to the spectral norm, which means that

$$|G\|_* = \sup_{\sigma_1(Q)\leq 1}\langle Q, G\rangle = \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T G\right). \qquad (34)$$

We let $G = U\Sigma V^T = \sum_{i=1}^{r}\sigma_i u_i v_i^T$ be the SVD of G, and define $Q = UV^T$. Then, $\sigma_1(Q) = 1$ by construction, and we have

$$
\begin{aligned}
\langle Q, G\rangle &= tr\left(Q^T G\right) \\
&= tr\left(VU^T U\Sigma V^T\right) = tr(\Sigma) = \|G\|_*. \qquad (35)
\end{aligned}
$$

Thus, we know that $|G\|_* \leq \sup_{\sigma_1(Q)\leq 1} tr(Q^T G)$.

Furthermore, the other direction can be proved as

$$
\begin{aligned}
&\sup_{\sigma_1(Q)\leq 1}\langle Q, G\rangle \\
&= \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T G\right) = \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T U\Sigma V^T\right) \\
&= \sup_{\sigma_1(Q)\leq 1}\langle U^T QV, \Sigma\rangle \\
&= \sup_{\sigma_1(Q)\leq 1}\sum_{i=1}^{r}\sigma_i(G)\left(U^T QV\right)_{ii} \\
&= \sup_{\sigma_1(Q)\leq 1}\sum_{i=1}^{r}\sigma_i(G)\left(P^T\lambda W\right)_{ii} \quad (do\ SVD\ on\ Q) \\
&= \sup_{\sigma_1(Q)\leq 1}\sum_{i=1}^{r}\sigma_i(G)\sigma_i(Q)p_i^T w_i \\
&\leq \sup_{\sigma_1(Q)\leq 1}\sum_{i=1}^{r}\sigma_i(G)\sigma_1(Q)\|p_i\|_2\|w_i\|_2 \\
&= \sum_{i=1}^{r}\sigma_i(G) = \|G\|_*. \qquad (36)
\end{aligned}
$$

Thus, we know that $|G\|_* = \sup_{\sigma_1(Q)\leq 1} tr(Q^T G)$. Finally, the subadditivity of the nuclear norm can be proved as follows:

$$
\begin{aligned}
\|A + B\|_* &= \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T(A + B)\right) \\
&\leq \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T A\right) + \sup_{\sigma_1(Q)\leq 1} tr\left(Q^T B\right) \\
&= \|A\|_* + \|B\|_*. \qquad (37)
\end{aligned}
$$

□

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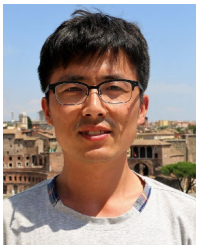
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